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Clustering

Story So Far

Given data points.

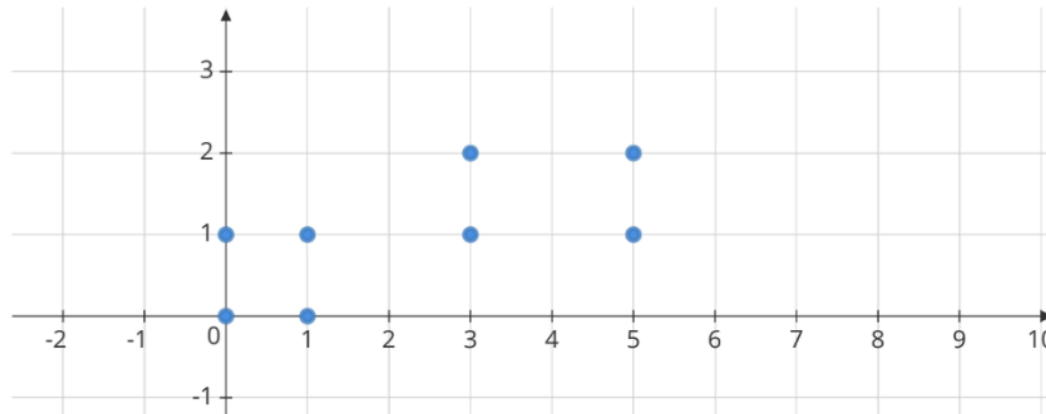
Find eigen directions with maximum variance.

Project proxies of the data points along the eigen direction.

But, by applying PCA on the given dataset Have we really understood the underlying structure in the data ? It is one thing to say that there is a linear subspace that has most of the information that is present in the data. **But within the linear subspace there there may be more structure to be uncovered.**

$$X = \begin{bmatrix} 0 & 0 & 1 & 1 & 3 & 3 & 5 & 5 \\ 0 & 1 & 0 & 1 & 1 & 2 & 1 & 2 \end{bmatrix}$$

The shape of the dataset is $d \times n$, where d is 2 and n is 8



Input

$$\{x_1, x_2, \dots, x_n\} \mid x \in \mathbb{R}^d$$

Goal

Partition the n datapoints into k different parts

Example

$$\{x_1, x_2, x_3, x_4, x_5\} \mid x_i \in \mathbb{R}^d$$

Partition the 5(n) datapoints into 3(k) different clusters

Possible Clusters

$$\begin{aligned} &\{x_1, x_5, x_3\} \{x_2\} \{x_4\} \\ &\{x_1, x_2\} \{x_3, x_4\} \{x_5\} \\ &\quad \cdot \\ &\quad \cdot \\ &\quad \cdot \\ &243 \ (3^5) \text{ possibilities} \end{aligned}$$

In general k^n cluster possibilities are there

Notations

Datapoints

$$x_1, x_2, \dots, x_n$$

Cluster Indicator

$$z_1, z_2, \dots, z_n$$

Each datapoint belongs to k cluster

$$z_i \in \{1, \dots, k\}$$

L1 Norm – Manhattan Norm

$$\|x\|_1 = \sum_{i=1}^n |x_i|$$

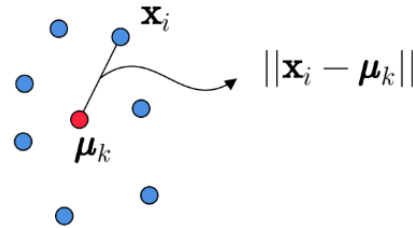
L2 Norm – Euclidean Norm

$$\|x\|_2 = \sqrt{\sum_{i=1}^n x_i^2}$$

Question

Given a cluster assignment how good is it ?

$$F(z_1, z_2, \dots, z_n) = \sum_{i=1}^n \|x_i - \mu_{z_i}\|_2^2$$



Indicator Function

$$\mathbf{1}(\text{cond}) = \begin{cases} 1, & \text{cond is true} \\ 0, & \text{cond is false} \end{cases}$$

How to Calculate Cluster Mean

for cluster k :

$$\mu_k \in \mathbb{R}^d$$
$$\mu_k = \frac{\sum_{i=1}^n x_i \cdot \mathbb{1}(z_i = k)}{\sum_{i=1}^n \mathbb{1}(z_i = k)}$$

Goal

$$\min \{z_1, \dots, z_n\} = \sum_{i=1}^n \|x_i - \mu_{z_i}\|_2^2$$

But too many possibilities ! (k^n)

Thus it is a NP Hard Problem

Lloyd's Algorithm (k - means)

For this K means problem, Lloyds algorithm is a solution

Initialisation – Randomly assign cluster indicator

$$z_1^0, z_2^0, \dots, z_n^0 \in \{1, \dots, k\}$$

Until convergence

1. Compute Means $\forall k$

$$\mu_k^t = \frac{\sum_{i=1}^n x_i \cdot 1(z_i^t = k)}{\sum_{i=1}^n 1(z_i^t = k)}$$

2. Reassignment Step $\forall i$

$$z_i^{t+1} = \arg \min_k \|x_i - \mu_k^t\|_2^2$$

If the mean of the current assignment is smallest, then donot re-assign

Fact: Llyods Algorithm Converges but the converged solution may not be the the optimal solution yet produces reasonable cluster in practice.

Convergence of K means Algorithm

Lets say we are at iteration t of Llyods algorithm.

Given,

current cluster assignments :

$$z_1^t, z_2^t, \dots, z_n^t \in \{1, \dots, k\}$$

mean of cluster k in assignment t

$$\mu_k^t$$

Say we update our assignment

$$z_1^{t+1}, z_2^{t+1}, \dots, z_n^{t+1} \in \{1, \dots, k\}$$

$$\sum_{i=1}^n \|x_i - \mu_{z_i^{t+1}}^t\|_2^2 < \sum_{i=1}^n \|x_i - \mu_{z_i^t}^t\|_2^2 \quad (1)$$

After the assignment has been made

$$\sum_{i=1}^n \|x_i - \mu_{z_i^{t+1}}^{t+1}\|_2^2 \leq \sum_{i=1}^n \|x_i - \mu_{z_i^t}^t\|_2^2 \quad (2)$$

from (1) and (2) we got

$$\sum_{i=1}^n \|x_i - \mu_{z_i^{t+1}}^{t+1}\|_2^2 < \sum_{i=1}^n \|x_i - \mu_{z_i^t}^t\|_2^2$$

This implies that the objective function strictly reduces after re-assignment

$$F(z_1^{t+1}, \dots, z_n^{t+1}) < F(z_1^t, \dots, z_n^t)$$

$\therefore$ There are only finite number of partitions (*at most* k^n) and once a partition has been visited, it cannot be revisited. so eventually the algorithm has to converge surely (may be to local minima)

Nature of Clusters Produced

Lets say $K = 2$

Llyods algorithm converges and produces two cluster with means μ_1 and μ_2

What can we say about points assigned to the cluster ?

By algorithm construction(for cluster 1)

$$\|x - \mu_1\|^2 \leq \|x - \mu_2\|^2$$

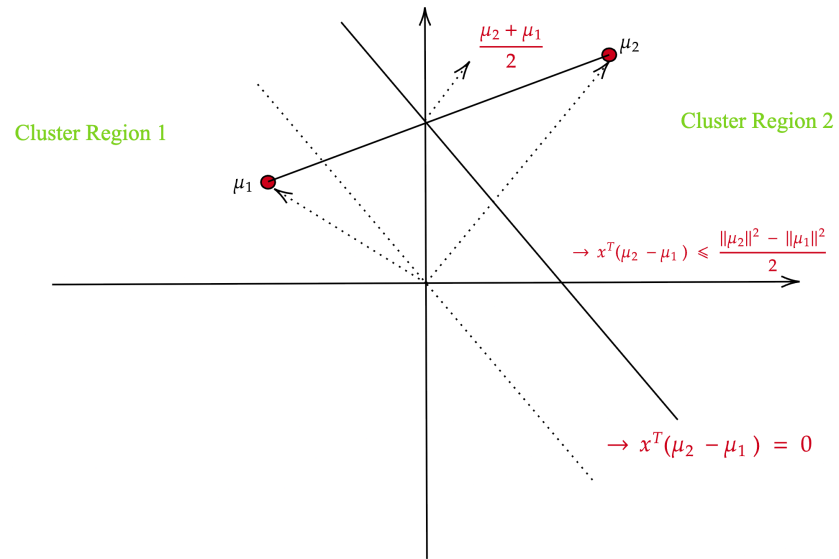
$$\|x\|^2 + \|\mu_1\|^2 - 2x^T \mu_1 \leq \|x\|^2 + \|\mu_2\|^2 - 2x^T \mu_2$$

$$-2x^T \mu_1 + 2x^T \mu_2 \leq \|\mu_2\|^2 - \|\mu_1\|^2$$

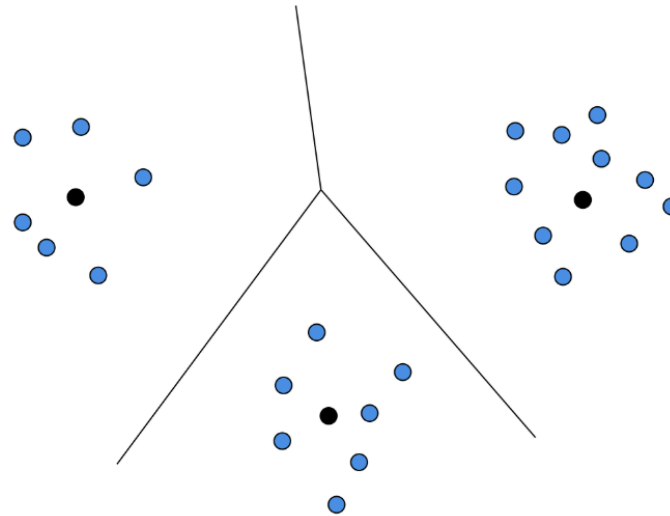
$$2x^T (\mu_2 - \mu_1) \leq \|\mu_2\|^2 - \|\mu_1\|^2$$

$$x^T (\mu_2 - \mu_1) \leq \frac{\|\mu_2\|^2 - \|\mu_1\|^2}{2}$$

$$x^T w \leq b$$



For $K = 3$



Cluster regions are intersection of half spaces known as voronoi region

Initialisation of centroids

- Pick k-means uniformly at random from the dataset
- K means ++

K-means++ is an algorithm that provides a sounder initialization which results in some convergence guarantees. The algorithm is probabilistic in nature. We will describe the algorithm for $k = 3$

Step-1: Choose any of the n data-points as the first mean μ_1 by sampling uniformly at random from $\{x_1, \dots, x_n\}$

$$P(x_1 = \mu_1) = \frac{1}{n}$$

Step-2: Choose the second mean as one of the remaining data-points using this process

$$d(x_i, \mu_1) = \text{distance of } x_i \text{ from mean } \mu_1$$

The score for each data-point is the squared distance:

$$s(x_i) = d(x_i, \mu_1)^2$$

A probability distribution over the n-1 points. Recall that the summation starts from 2 since x_1 has been chosen.

$$p(x_i) = \frac{s(x_i)}{\sum_{j=2}^n s(x_j)}$$

Sample a data-point using this distribution and assign it to μ_2 . For convenience, relabel the sampled point x_2 .

That is, if x_5 is sampled, swap the labels and the scores for x_2 and x_5

$$p(\mu_2 = x_2 \mid \mu_1 = x_1) = \frac{s(x_2)}{\sum_{j=2}^n s(x_j)}$$

Step-3: Choose the third mean as one of the remaining data-points using this process

$$d(x_i, \mu_j) = \text{distance of datapoint } x_i \text{ from mean } \mu_j$$

The score for each data-point is the squared distance of the distance of x_i to the closest mean :

$$s(x_i) = \min(d(x_i, \mu_1), d(x_i, \mu_2))^2$$

A probability distribution over the $n-2$ points. Recall that the summation starts from 3 since x_1 and x_2 has been chosen.

$$p(x_i) = \frac{s(x_i)}{\sum_{j=3}^n s(x_j)}$$

Sample a data-point using this distribution and assign it to μ_3 . For convenience, relabel the sampled point x_3 .

That is, if x_7 is sampled, swap the labels and the scores for x_3 and x_7

$$p(\mu_3 = x_3 \mid \mu_2 = x_2, \mu_1 = x_1) = \frac{s(x_3)}{\sum_{j=3}^n s(x_j)}$$

The overall probability of choosing these three means in this order is the probability of the three probabilities we have computed so far. For $K > 3$, the algorithm can be extended in a similar fashion.

k-means ++ is a computationally heavier but it is practically a solid way to initialize lloyds algorithm.

Choice of K

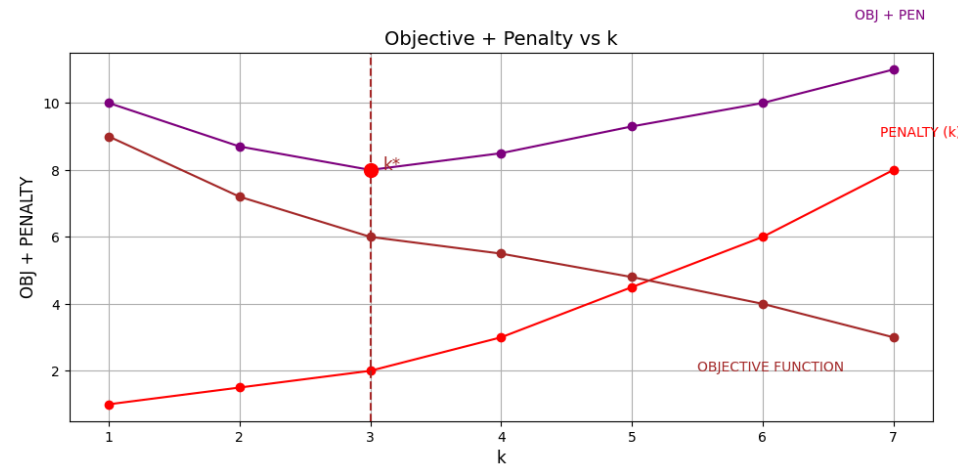
k is a hyperparameter and must be chosen appropriately. A hyperparameter is different from a parameter in that it is not "learnt from data" but is chosen before the learning begins. One heuristic to choose a value of k is the elbow method.

The elbow method selects the optimal number of clusters k by plotting the within-cluster sum of squares (WCSS) against k and identifying the point where the rate of decrease sharply slows, forming an "elbow."

when is the objective function optimal?

$$F(z_1, z_2, \dots, z_n) = \sum_{i=1}^n \|x_i - \mu_{z_i}\|_2^2$$

We want k to be as small as possible, thus we penalize large values of k



Some Common Criterion As Well

Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) are used in K-means clustering to select the optimal number of clusters k by balancing model fit and complexity. Both are based on the log-likelihood of the data under a Gaussian mixture assumption and include a penalty term to discourage overfitting as k increases. AIC penalizes complexity with a constant factor, while BIC applies a stronger penalty based on the number of data points. In K-means, for each k , we compute the within-cluster sum of squares (WCSS), estimate the likelihood, and calculate AIC and BIC scores. The optimal k is the one that minimizes these criteria, providing a statistically grounded alternative to heuristic methods like the elbow method.

Application of K-means

The k-means algorithm has a wide range of applications across various domains.

- One of its primary uses is in customer segmentation, where companies divide their customers into groups based on common characteristics to target specific clusters for advertising campaigns.
- Another application is in document classification, where the algorithm is used to allocate various classes or categories to documents.
- Additionally, k-means clustering is used in recommendation engines, where it helps in recommending products or services to users based on their past purchases or preferences.
- In the field of image processing, k-means is applied for image segmentation, which involves dividing an image into different segments to identify specific objects, boundaries, or patterns.
- The algorithm is also utilized in the analysis of data sets, such as in the case of the Breast Cancer Wisconsin Diagnosis dataset, where it is used for classifying breast tumors as malignant or benign.

Reference

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Karthik Thiagarajan, Machine Learning Techniques Github Notes.

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Stanford University, CS229 Machine Learning Notes

Thank You !